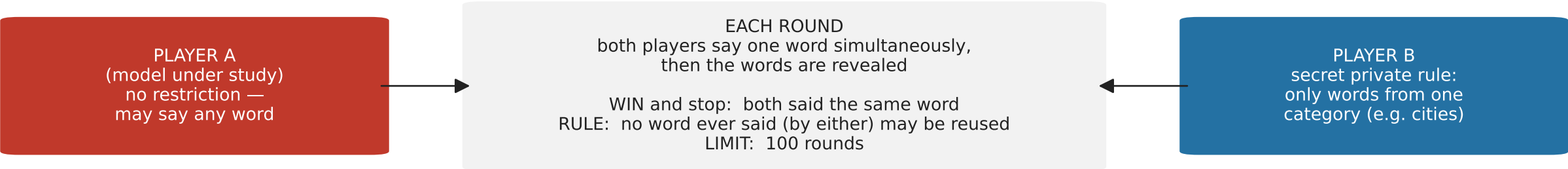


Game 1: Simultaneous word-convergence with a hidden category



, A must infer B's hidden category from B's words and say category words. Qwen3-32B (both roles): 14/24 games meet within 100 rounds (median 8 rounds when they do); 10/24 hit the

Typical failure mode (game 10): the word-family loop

round	Player A	Player B
5	ranch	vienna
6	vivid	oslo
7	ripe	havana
8	virus	dubai
9	dread	seville
10	dared	geneva
11	dare	athens
12	ripen	warsaw
13	ripest	munich
14	riper	brussels
15	ripeness	zurich
16	riperest	stockholm
17	ripests	prague
18	riperly	reykjav
19	riperer	lisbon
20	riperers	cape
21	riperess	moscow
22	riperity	bangkok
23	ripered	accra
24	riperely	melbourne
25	riperness	helsinki
26	riperes	nairobi
	⋮	⋮

A locks into the “ripe-” word family for 36 consecutive rounds, inventing non-words to satisfy the no-repeat rule, while B produces a city every round. (This game eventually escaped the loop and met at round 97.)

Success case (game 4): category adoption

round	Player A	Player B
1	ballet	paris
2	tango	tokyo
3	salsa	rio
4	samba	madrid
5	flamenco	buenos
6	bolero	seville
7	seoul	new
8	waltz	vienna
9	jazz	kiev
10	beijing	brussels
11	chachacha	moscow
12	budapest	rome
13	athens	bucharest
14	geneva	oslo
15	bern	berlin
16	prague	prague

MEET

A starts with dances, then adopts B's evident city category (highlighted) and the players meet on “prague” at round 16.